

Def. A linear operator $E: V \rightarrow V$ is called a projection if $E^2 = E$.

Lemma. Let $E: V \rightarrow V$ be a projection. Then,

1) $\text{Im } E = \{x \in V \mid E(x) = x\}$,

2) $V = \text{Im } E \oplus \ker E$

Proof. $x \in \text{Im } E \Leftrightarrow x = E(\alpha)$ for some $\alpha \in V \Rightarrow E(x) = E^2(\alpha) = E(\alpha) = x$. The converse is trivial!

Given $x \in V$, $x = E(x) + (x - E(x))$. Clearly $E(x) \in \text{Im } E$ and

$$E(x - E(x)) = E(x) - E^2(x) = (E - E^2)(x) = 0, \text{ so } x - E(x) \in \ker E.$$

And, $E(y) = y$ and $E(y) = 0$, then $y = 0$. Thus $\text{Im } E \cap \ker E = \{0\}$.

Thm. Let $V = W_1 \oplus W_2$. Then, there is one and only one projection $E: V \rightarrow V$ s.t.

$$\text{Im } E = W_1 \text{ and } \ker E = W_2$$

Proof. Let E be a projection. For β_1 , basis for $\text{Im } E$, and β_2 , basis for $\ker E$,

$$[E]_{\beta_1 \cup \beta_2} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \text{ where } r = \text{rank } E.$$

Comment: In General,

let $\{\alpha_1, \dots, \alpha_k\}$ be a basis for $\text{Im } E$ and $\{\alpha_{k+1}, \dots, \alpha_n\}$ be a basis for $\ker E$.

Then, the invertible Matrix $A = \begin{bmatrix} \alpha_1 & \alpha_2 & \dots & \alpha_n \end{bmatrix}$ satisfies

$$EA = \begin{bmatrix} E\alpha_1 & \dots & E\alpha_n \end{bmatrix} = \begin{bmatrix} \alpha_1 & \dots & \alpha_k & 0 & \dots & 0 \end{bmatrix}$$

Therefore, $E = \begin{bmatrix} \alpha_1 & \dots & \alpha_k & 0 & \dots & 0 \end{bmatrix} \cdot A^{-1} = A \cdot \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} \cdot A^{-1}$ is we desired.

Thm. Let V be a fin. dim. v.s over F .

If $V = \bigoplus_{i=1}^k W_i$, then there exist k linear operators E_i on V s.t.

$$E_i \circ E_j = \delta_{ij} \cdot E_j, \quad I = \sum_{i=1}^k E_i, \quad W_i = \text{Im } E_i \quad (1 \leq i \leq k) \quad \dots (*)$$

Conversely, if $E_1, \dots, E_k$ are k linear operators on V which satisfy $(*)$, then

$$V = \bigoplus_{i=1}^k W_i$$

Proof. Suppose $V = \bigoplus_{i=1}^k W_i$. For each $1 \leq i \leq k$, define $E_i: V \rightarrow V: \sum_{i=1}^k \alpha_i \mapsto \alpha_i$. proved.

Conversely, suppose that $E_1, \dots, E_k$ are linear operators on V satisfying $(*)$.

Given $\alpha \in V$, $\alpha = E_1(\alpha) + \dots + E_k(\alpha) \in W_1 + \dots + W_k$. And, to show uniqueness such expression,

let $\alpha = \alpha_1 + \dots + \alpha_n$, where $\alpha_i \in W_i$. And, let $\alpha_i = E_i(\beta_i)$, for each $1 \leq i \leq k$.

Now, for each j , $1 \leq j \leq k$,

$$E_j(\alpha) = \sum_{i=1}^k E_j(\alpha_i) = \sum_{i=1}^k (E_j \circ E_i)(\beta_i) = E_j^2(\beta_j) = E_j(\beta_j) = \alpha_j$$

Thus proved.